Magnetic Force Enabled Plant Seed Levitation to Simulate Microgravity Environment on A Microfluidic Chip

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Abstract— **For plants on Earth, gravitropism is essential element affecting seedling establishment and plant development. As an integrated part of plant growth and development, gravitropic response is mainly mediated by alternation of photohormone auxin. In microgravity environment, the study of auxin responses is critical to illustrate how plants perceive environmental gravity conditions and then initiate auxin signaling pathway. Nevertheless, the major challenge is the lack of a tool that can provide a simulated microgravity condition. In this study, we developed a microfluidic negative-magnetophoresis (MNM) platform to levitate Arabidopsis seeds in an equilibrium plane where the applied magnetic force compensates gravitational acceleration. With the advantage of the MNM platform to mimic a microgravity environment on chip, it's found that auxin response is significantly repressed during seed levitation. The MNM can enable the study of physiological process and molecular mechanism of hormone in a simulated microgravity condition, which benefits further applications for plant space biology.**

I. Introduction

Gravity is unique and constant on Earth throughout the entire life cycle. Plants are sessile and well adapted to this 1g level to form gravitropsim, which is essential for plant to develop growth pattern and establish evolutionary adaptations to stimuli, e.g. light, water and mineral salts [1]. Hence, the alteration of gravity profoundly influences physiological processes and initiates adaptive responses. As a cardinal regulator of all aspects of plant growth and development [2], auxin has been revealed to involve in response to gravity stimulus through redistribution of auxin gradients [3]. Therefore, it's important to study the role of auxin during plant development in microgravity condition, which is essential for plant space biology.

In general, the way to encounter microgravity environments is merely in outer space or during free fall. Nevertheless, continuous and long-term microgravity is still a challenge, as the International Space Station (ISS), Space Shuttle and free fall tower merely provide costly and short-term microgravity condition [4]. More importantly, the above-mentioned methods are inapplicable for routine laboratory research. As alternatives, devices that simulate microgravity have been developed to meet such challenges. Random Positioning Machine (RPM) as a typical simulator can mimic microgravity through rotary system, such as 1D-clinostat, 2D-clinostat and 3D-clinostat. Although RPM can generate comparable effects of microgravity when directional changing is fast enough, the uniformity of gravity and *in situ* mechanical stress introduction remain unsolved [5].

Magnetic levitation technique has also been utilized to simulate microgravity via either positive or negative magnetophoresis [6,7]. However, positive magnetophoresis levitates the subject of interest by the application of magnetic nanoparticles that are loaded inside of the subject or combined to its surface [8]. Conversely, negative magnetophoresis-mediated levitation can exactly mimic microgravity through counteracting gravitational force in a manner of free labeling. In this technique, objects are immersed in a magnetic solution which locates in a magnetic field. Due to magnetic force, the paramagnetic materials suspended in the solution can generate a concentration difference over the magnetic field area, the higher the magnetic field gradient, the higher the concentration of the paramagnetic materials. Therefore, the nonmagnetic objects are triggered moving towards regions of low magnetic field along magnetic field gradients, resulting in stable magnetic levitation and simulation of microgravity condition [9]. In this method, gadolinium (Gd^{3+}) based solution has been utilized as the magnetic solution, due to its good biocompatibility, high efficiency and comparably inexpensive [10-12].

In this study, we used negative-magnetophoresis in a microfluidic chip to establish a magnetic levitation system for *in situ* plant seed culture. The effects of a various concentrations of Gd^{3+} on seed levitation are assessed, and the auxin responses to Gd^{3+}-triggered microgravity are also investigated. The development of such platform has led to better understanding of not only the microgravity phenomenon but also its application to diverse biological samples in future.

II. Experimental

A. Levitation Device Fabrication

The microfluidic device was realized by plasma bonding between a polydimethylsiloxane (PDMS) with microfluidic structures and a glass substrate. The PDMS layer was replicated from photoresist structures that were fabricated

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by soft lithography. In brief, a 180 μm thick layer of SU8 2100 (19060515, MicroChem Corp.) was spin-coated onto a silicon wafer and soft baked at 95 ℃ for 35 minutes. The coated wafer was then exposed to the photomask containing the chip design at 280 mJ/cm^2 using the EVG mask aligner (610TB, EV Group GmbH). After 13 minutes' post exposure bake at 95 ℃, the wafer was then placed in a dish of SU-8 developer for approximately 12 minutes, gently agitated until the exposed features remain. The wafer was rinsed with clean IPA and then dried with N_2 gas. After master mold was fabricated, a 10:1 mixture of PDMS prepolymer and curing agent (Sylgard 184, Dow Corning Corp.) was cast over the wafer and cured at 80 ℃ overnight. Finally, the PDMS replica was peeled off and sealed by a glass substrate using plasma bonding.

B. Simulation

Governing equations

- Magnetic force

The magnetic force F_m is given by Eq. (1):

$$F_m=\frac{V_s(\chi_s-\chi_m)}{\mu_0}(B\cdot\nabla)B \quad (1)$$

where μ_0 is the permeability of free space, V_s is the volume of seed, χ_s and χ_m are the magnetic susceptibility of the seed and Gd^{3+} solution (381667, Sigma), respectively. The relationship between magnetic susceptibility and paramagnetic salt concentration is given by Eq. (2) [13]:

$$\chi_m=(0.35\times C_{Gd}-9.04)\times10^{-6} \quad (2)$$

where C_{Gd} is the concentration of Gd^{3+} solution (mM).

- Hydrodynamic drag force

The hydrodynamic drag force F_d is given by Eq. (3):

$$F_d=6\pi\eta_s r\Delta v f_D \quad (3)$$

where r is the radius of particle (here the semi-major axis of the ellipsoid is instead of r), Δv is the velocity difference between the seed and the Gd^{3+} solution, η_s is the viscosity of the Gd^{3+} solution, f_D is the drag coefficient of the particle and incorporates the influence of a solid wall in the vicinity of the moving seed, given by Eq. (4):

$$f_D=\left[1-\frac{9}{16}\left(\frac{r}{r+z}\right)+\frac{1}{8}\left(\frac{r}{r+z}\right)^3-\frac{45}{256}\left(\frac{r}{r+z}\right)^4-\frac{1}{16}\left(\frac{r}{r+z}\right)^5\right]^{-1} \quad (4)$$

z is the distance of the seed to the solid wall.

- Viscous resistance

The gravity F_g is given by Eq. (5):

$$F_g=m_s g \quad (5)$$

where m_s is the weight of a single seed and g is the gravitational constant.

- Buoyancy

The buoyancy F_b is given by Eq. (6):

$$F_b=\rho_m g V_s \quad (6)$$

where ρ_m is the density of Gd^{3+} solution.

Based on the governing equations above, we used COMSOL software to simulate the movement of the seed in Gd^{3+} solution.

C. Plant Growth and Seed Harvest

The Arabidopsis wild-type and transgenic plants used in this study were in a Col-0 (Columbia) background, ordered from Nottingham Arabidopsis Stock Centre (NASC). All plants were grown in a controlled LED-light grow chamber (MGC-100HP-2L, Shanghai Yiheng Technical Co., Ltd.) at 20-22 ℃ under cool-white fluorescent light (80-100 μmol/m^{2}s) in a long-day photoperiod (16 h light/8 h dark). Arabidopsis seeds were harvested only after the siliques have completely browned.

D. Seed Stratification and Germination

Surface of the Arabidopsis seeds were sterilized using 50% (v/v) bleach solution in tubes [14]. After sterilization, seeds were suspended in autoclaved distilled water and then kept at 4 ℃ in refrigerator for 2 days to break seed dormancy. After refrigerated stratification, seeds were sowed in cylindrical channels (1000 μm in diameter) on chip, then cultured and germinated with half-strength Murashige and Skoog (1/2 MS) medium for 1 day.

E. Cryostat Slicing and fluorescent labeling

The seeds were embedded in Optimal Cutting Temperature compound (OCT, 23-730-571, Fisher HealthCare) in a container. Then, OCT-embedded seeds were snap frozen and stored in the container at -80 ℃ in refrigerator for at least 2 hours. Sliced sections (25 μm) were cut from one side of the seeds to the opposite side continuously using a cryostat (CM1950, Leica). From approximately 20 slides per seed, each slide was placed on a glass slide and dried at room temperature. These slides were then washed with phosphate buffered saline (PBS, pH7.4) buffer to remove OCT and fixed with paraformaldehyde (PFA, 158127, Sigma) for 10 minutes. After washing in PBS buffer again, the slices were mounted with DAPI solution (10236276001, sigma) and sealed with cover glasses.

III. RESULTS

The chip is consisted of 5 straight microchannels which are arranged in parallel, each microchannel has 2 cultivation reservoirs to independently control the levitation process of a single seed in a reservoir. Thus, a single chip enables simultaneous operation of 10 seeds (see Fig. 1A). Below the glass substrate, there are 2 NdFeB permanent magnets (N52, 1" × 1/4" × 1/4") in opposite polarities providing a high-gradient magnetic field (see Fig. 1B). The distance between two permanent magnets is 1000 μm, and the thickness of glass substrate is 200 μm. The microfluidic device for seed levitation fabricated by soft lithography is shown in Fig. 1C.

The processes of seed culturing and Gd^{3+}-mediated levitation are illustrated in Fig. 1D. Firstly, the Arabidopsis seeds are put into the cultivation reservoirs, the culture medium is injected into the microchannels by using a neMESYS syringe pump, the seeds are placed at the bottom of the holes and are cultured for 24 hours. On the 2nd day, 200 mM Gd^{3+} in culture medium is introduced into the cultivation holes through the microchannels to replace the 1/2 MS medium. Once the microchannel is fulfilled with Gd^{3+} solution, seeds are observed to gradually levitate to an equilibrium position in each cultivation reservoir. After maintaining the seeds at the equilibrium position for 30 minutes, the Gd^{3+} solution is then replaced by a mixed

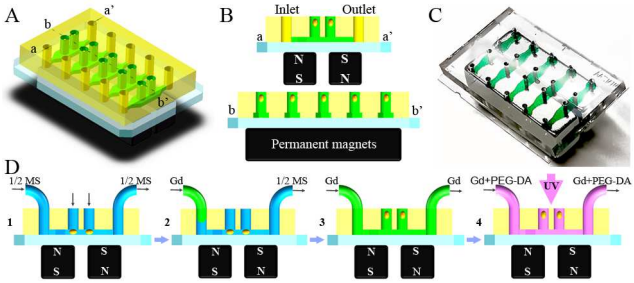

Figure 1. Chip design and experiment procedures. (A) Photo of the microfluidic chip for seed levitation. (B) Design of the chip with 5 micro-channels for seed cultivation, each channel (10 mm in length, 2 mm in width and 100 μm in hight) has two cultivation reserviors (1000 μm in diameter). (C) Layout of the microdevice showing the position of the magnets and the plant seeds. (D) The experimental procedures. First, the seeds are put into the cultivation reservoirs and the culture medium is injected into microchannels. Second, the culture medium is replaced by Gd^{3+} solution. Third, the seeds are lifted and maintained at the position by negative magnetophoresis. Last, the PEG-DA and Gd^{3+} solutions are injected into the channels and photopolymerized by ultraviolet radiation.

solution of Gd^{3+} and 6% (v/v) poly(ethylene glycol)-diacrylate (PEG-DA) 575, and the chip is irradiated with ultraviolet rays to solidify the mixed solution, thus levitated seeds are immobilized in the solution for subsequent slicing process.

In order to cultivate the seeds in the best suspension state, we established a simulation model of a seed in the magnetic solution. COMSOL software was used to qualitatively analyze the force and suspension trajectory under the action of the permanent magnet. Accordingly, the position of the cultivation reservoirs was optimized (see Fig. 2A). Here the seed was simplified to an ellipsoidal (a = 0.25 mm, b = 0.175 mm, c = 0.175 mm), the initial vertical height of its center was 0.2 mm (Y = 0.2 mm), and the initial horizontal position was changed from 0 to 2.5 mm. The seeds in the Gd^{3+} solution were subjected to the magnetic force F_m, hydrodynamic drag force F_d, gravity F_g and buoyancy F_b (see Fig. 2B). The magnetic force was the main power to levitate the seed, which is given by Eq. (1). The main factors that determine the magnitude of the magnetic force are the magnetic field strength and the magnetic field gradient. The simulation results show that near the edges of the two magnets, both the magnetic field strength and the magnetic field gradient reach the maximum values, so the magnetic force near the magnet edge (X = ± 0.5 mm) is greater than that in other positions, and the height of the seeds in the Y direction is also higher, however, the magnetic force and the height of the suspension are very small in the position away from the edge of the magnets (X = ± 2.5 mm), as shown in Fig. 2C. The simulation result indicates that the area above the two magnet edges is the most suitable for seed suspension, so the cultivation reservoirs are designed along and above the magnet edges (see Fig. 1).

Before initiation of Gd^{3+}-promoted seed levitation, the toxicity of Gd^{3+} solution on seed germination was tested. The seeds, i.e. immersed in Gd^{3+} solution for 30 minutes, represented slightly difference from those untreated seeds during subsequent germination process (see Fig. 3A). Thus,

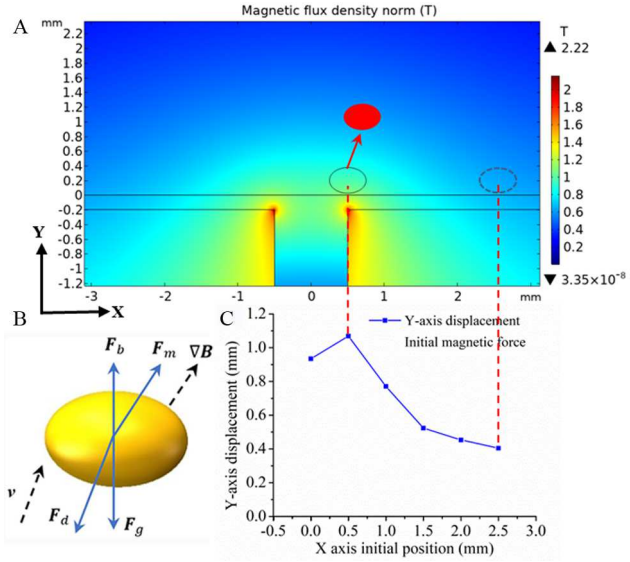

Figure 2. System optimization for seed levitation. (A) The simulation of seed levitation using COMSOL Multiphysics. (B) The force analysis of seed in Gd^{3+} solution. (C) The simulation results of initial magnetic forces and Y-axis displacements of seeds at different X-axis positions.

200 mM Gd^{3+} is a safe condition for plant seeds during short period of stimulated levitation. To shape the ellipsoid-like seeds, the parameters a, b and c (the lengths along x, y, z axis) are measured and calculated according to microscope images. After statistical analysis, the lengths of a, b and c are 578.0 ± 60.6 μm, 406.1 ± 30.5 μm, 335.6 ± 40.7 μm, respectively. To ensure Gd^{3+}-promoted seed levitation in solution, the half of the longest axis of seeds (~ 300 μm) is the levitated threshold, i.e. the levitation distance should be longer than 300 μm. It was observed that seed floating can be guaranteed at the concentration of 200 mM, as the levitation distance was over 600 μm (see Fig. 3B). Thus, 200 mM Gd^{3+} is a safe and sufficient condition to create a simulated microgravity niche for Arabidopsis seeds on chip.

Auxin is an essential phytohormone that contributes to all aspects of the plant development. It is revealed that plant endogenous level of auxin in microgravity environment is nearly comparable to that under 1g condition on the whole [15]. However, cellular distribution of auxin is extremely lacking. Here, we study the auxin response of plant embryo on cellular level using well-established indicator DR5::GFP (see Fig. 4). Before the examination of auxin response to the simulated microgravity, auxin responses of embryo to Gd^{3+} solution were examined. As shown in Fig. 4B, short-term treatment of Gd^{3+} hardly affected auxin signal during early germination. Through application of magnets under Gd^{3+} solution, seeds were levitated and then kept for 30 minutes, auxin response was dramatically repressed after seed levitation (see Fig. 4C). Auxin responses are the downstream reflection of spatiotemporal distribution of auxin, depending on not only its metabolism but also its directional cell-to-cell transportation [2]. In the future, the studies of magnetic levitation on auxin synthesis and transportation signaling pathway will help to illustrate underlined mechanism of auxin response to microgravity.

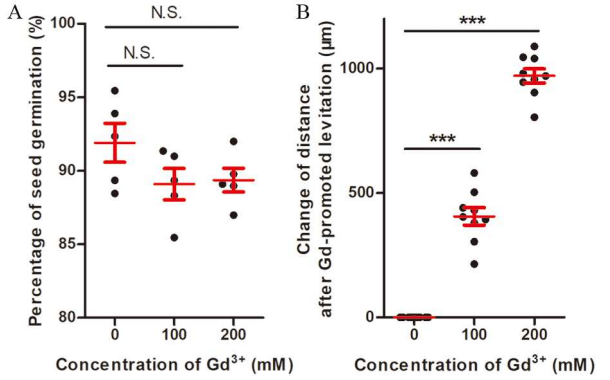

Figure 3. Gd^{3+}-promoted magnetic levitation of Arabidopsis seeds *in situ* on the chip. (A) Quantification on the percentage of seed germination after treatment by the indicated Gd^{3+} solutions for 30 minutes. Data are mean values ± standard error (SE), with six biological repeats for each bunch of Arabidopsis seeds (n=30). (B) Histograms on the levitation height change of the seeds at indicated Gd^{3+} solutions. Dots represent mean values ± SE from ten seeds on-chip.

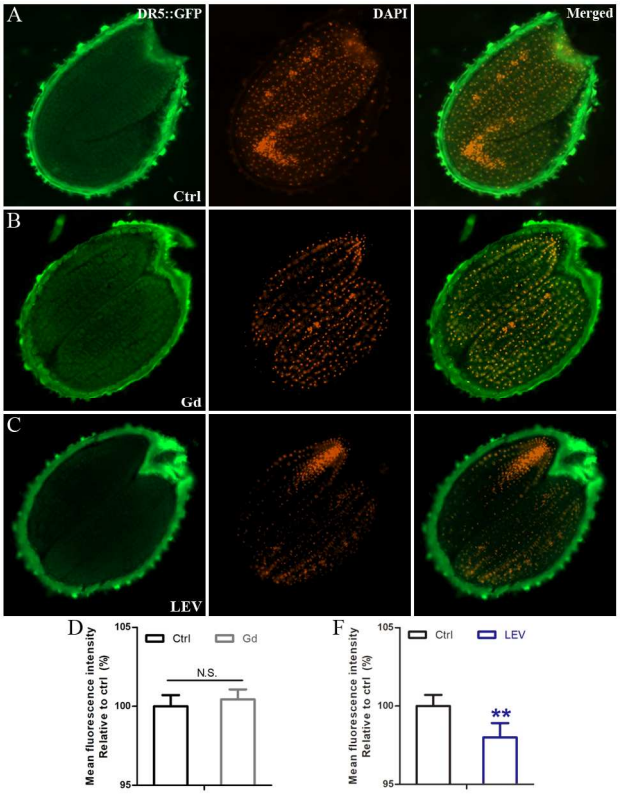

Figure 4. The auxin response of Arabidopsis seeds during magnetic levitation. (A-C) Expression of DR5::GFP in embryo sections of Arabidopsis seeds without Gd^{3+} solution(A), with Gd^{3+} solution only (B), and with Gd^{3+}-mediated magnetic levitation (C). The green color is GFP and the oval outline in strong green is auto-fluorescence from seed coats. The orange is fluorescence from 4',6-diamidino-2-phenylindole (DAPI) dye. Scale bar = 100 μm. (D-F) Quantification on the relative expression of DR5::GFP in embryo sections of Arabidopsis seeds with Gd^{3+} solution only (D) and with Gd^{3+}-mediated levitation (F). Error bars represent SE of the mean (n>30) with three biological repeats. Asterisks indicate significant differences (Student's t-test, **, P≤ 0.01).

IV. Conclusion

Auxin has been established as one of the critical factors in mediating plant gravitropic responses. However, due to the difficulties to generate microgravity condition on Earth, the existed microgravity simulators in lab cannot provide a spatio-temporal observation of plant on the cellular level. We designed and fabricated a microfluidic chip for long-term culturing of Arabidopsis seeds. With the application of Gd^{3+}-promoted negative magnetoresistance, plant seeds were levitated on chip to simulate microgravity condition. In the meantime, auxin response of plant embryo was found with an obvious decrease. In future, the mechanism of auxin response to microgravity in plant can be further studied with this microfluidic device. Furthermore, due to the convenience and straightforwardness of this approach, the MNM platform will be widely applied to a variety of biological samples.

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